

Improving Tip-of-the-Tongue Retrieval with Transformer-Based Reranking

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Abstract—Conventional information retrieval systems are mainly created to respond to short keyword-based searches. Nevertheless, numerous real-life search cases include vague, descriptive, and memory-based queries in which users are not in a position to remember specific words. This challenge is reflected in the TREC Tip-of-the-Tongue 2024 task which asks systems to recall pertinent documents when given long narrative queries. The proposed project examines the hypothesis of whether reranking using transformers can enhance retrieval compared to a lexical baseline. Early examination indicates that the query and document lengths are huge when compared to the normal levels of retrieval and the lexical matching is a problematic condition. This encourages the application of semantic models with a view to ensuring a superior contextual meaning.

I. INTRODUCTION

Traditionally, the information retrieval (IR) systems used lexical matching methods like BM25. Although they are useful in keyword search, they have difficulties in cases where the user is unable to remember the specific terms and instead enters descriptive or rough information. This situation is usually known as the “Tip-of-the-Tongue problem.”

The TREC Tip-of-the-Tongue 2024 dataset is a realistic and difficult retrieval task with long queries, narratives describing partially remembered objects (e.g. movies, landmarks in history or a famous person). Such queries are usually indirectly hinted, contain irrelevant information and paraphrased descriptions, so there is little lexical overlap between queries and relevant documents.

The proposed project will try to deal with this issue by examining whether reranking through transformers can enhance retrieval. The main hypothesis is that semantic models are suitable to understand the meaning of descriptive queries than purely lexical approaches.

II. DATASET AND EXPLORATORY ANALYSIS

The collection that was employed in this research is the TREC Tip-of-the-Tongue 2024 collection that included 3,185,450 documents and 600 test queries. The queries in this dataset are long, narrative descriptions of partially remembered entities unlike the traditional benchmarks of information retrieval. Such descriptions tend to contain indirect statements, personal impressions, and unnecessary information, which complicates the process of retrieving this information.

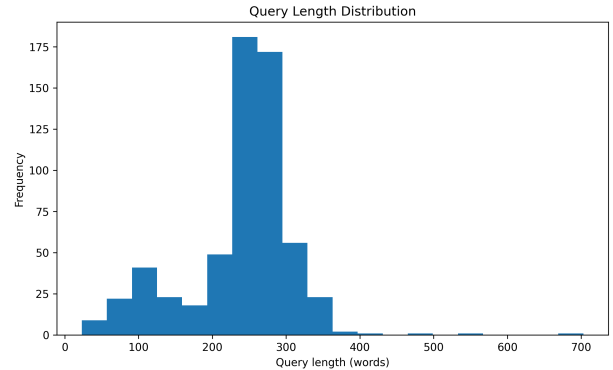


Fig. 1. Distribution of query lengths in the Tip-of-the-Tongue 2024 dataset.

Through exploratory data analysis it can be seen that the average length of query is 238.42 words with median length of 252 words and a maximum length of 703 words. It means that queries are much longer than in traditional IR tasks, where a query usually has a number of keywords. The distribution of query lengths is shown in Fig. 1.

The large range of query length indicates that users give different amounts of details, which could contain both useful and useless information. This increases the difficulty of identifying the most informative parts of the query.

Besides this, a sample of 5,000 documents demonstrates the mean length of a document of 6745.29 words, with a wide range of variation throughout the corpus. The distribution of document lengths is shown in Fig. 2.

The long queries and long documents combine to make the presentation of a double challenge. To begin with, lexical retrieval could be ineffective because of vocabulary differences between descriptive queries and formal document texts. Second, the longer papers are the more chances there are irrelevant information and thus harder to rank accurately.

To verify the quality of the dataset, a quality check was done to ensure that all the queries are valid in the dataset to be evaluated during retrieval. In general, the nature of the dataset reveals the necessity to use retrieval models able to retrieve semantically associated information beyond the direct matches on keywords.

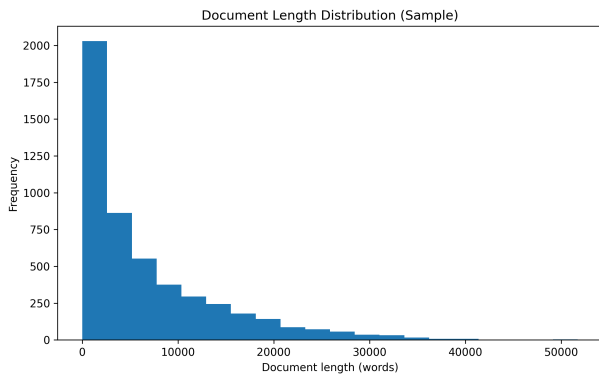


Fig. 2. Distribution of document lengths based on a sample of 5,000 documents.

III. RESEARCH QUESTIONS

This project is guided by the following research questions:

- Does transformer-based reranking improve retrieval effectiveness over a BM25 baseline for Tip-of-the-Tongue queries?
- How effective is semantic reranking for handling long and low keyword overlap queries?

These questions aim to evaluate the limitations of lexical retrieval and the potential benefits of semantic models in challenging IR scenarios.

IV. METHODOLOGY

The retrieval architecture that will be applied in the proposed system will be two-stage retrieval architecture, which is a common retrieval architecture in contemporary information retrieval systems. The first stage will use BM25 as a lexical retriever to effectively create a candidate set of documents. BM25 is selected as a baseline since it has high performance and efficiency in conventional retrieval environment.

BM25 however is dependent on word matching and can be ineffective when the query is using vocabulary that is different to those that are relevant. This drawback is especially significant in the Tip-of-the-Tongue context, in which questions are lengthy, descriptive, and indirect.

To overcome this shortcoming, the second step will use a transformer-based reranking model to narrow down the ranking of the retrieved documents. The transformer models can capture contextual and semantic relationship between queries and documents, and hence can discover relevant content where lexical overlap is low.

Mean Reciprocal Rank (MRR) and NDCG@10 will be used to evaluate the evaluation. Measurements of MRR will be the rank of the first relevant document in the ranking and NDCG@10 will be used to measure the quality of the ranking in the top results.

V. LITERATURE REVIEW

Transformer-based models have played a significant role in bringing about the new developments that have been made

in information retrieval recently. BERT [1] introduced deep bidirectional transformers added contextual understanding in language understanding models and allowed them to learn semantic associations other than matching keywords.

Nogueira and Cho [2] demonstrated that BERT can be effectively used for passage reranking. Their experiment revealed that BM25 combined with BERT-based reranking is much more effective in the retrieval process, especially in the situations when the semantic interpretation is needed.

Karpukhin et al. [3] proposed Dense Passage Retrieval, which uses dense vector representations to retrieve semantically relevant documents. The method will overcome the weaknesses of lexical retrieval by retrieving greater contextual similarities between queries and documents.

The general indication of all these studies is that lexical retrieval in conjunction with semantic models is a viable strategy in dealing with complex retrieval like the Tip-of-the-Tongue queries.

VI. PROJECT REPOSITORY

The project has been started via a public Git repository to provide an opportunity to re-create the project and develop it in a structured manner. There are also scripts of dataset inspection and exploratory data analysis and directory of the output, figure, and report.

The existing code includes the code to load the TREC Tip-of-the-Tongue dataset, to compute descriptive statistics, and to visualize distribution of query and document length. The subsequent stages of the project will further expand this repository to have BM25 indexing, retrieval experiment, and transformer models of reranking.

The repository is available at:

<https://github.com/ilaydaergur/DI725-assignment1.git>

VII. CONCLUSION

This project discusses how the Tip-of-the-Tongue retrieval can be applied to transformer-based reranking. The initial review indicates that the data is very challenging because of descriptive queries and lengthy documents. These features render lexical retrieval inadequate in its own right.

The suggested two-step retrieval system would overcome these difficulties by using both BM25 and transformer based reranking. The next action will be working on this pipeline and evaluating its effectiveness to conclude about the efficiency of semantic models in this context.

REFERENCES

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